

LAB 3 — COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

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Problem statement: Blood Bank Emergency Matcher

Problem / Scenario

The system supports emergency blood requests, blood availability checking, blood bank inventory management, request-status

updates, and emergency donor matching when the required blood is unavailable or insufficient.

2. Architecture Selection

Architecture Selection: We chose Layered Architecture for the Blood Bank Emergency Matcher System.

Two scenario-related reasons:

1. The Emergency Requester, Blood Bank Manager, and Donor interactions can be handled through separate user-facing components, while request processing, inventory management, and donor matching are handled by business components.
2. Blood requests, inventory records, donor information, and request status can be stored and managed separately in the data layer, making the system easier to organize and maintain.

Security Advantage:

The business layer can validate requests and authorize users before protected blood-bank, inventory, or donor information is accessed or modified.

Performance Benefit:

Separating request processing, availability checking, inventory operations, and donor matching into focused

components reduces unnecessary processing and keeps each operation close to the component responsible for it.

3. Components Identified

The main components are: Emergency Requester UI, Blood Request Management, Blood Bank Manager UI, Inventory Management, Donor UI, Emergency Donor Matching, Blood Bank Database, and Notification Service.

These components support requester interaction, inventory operations, donor matching, data storage and status notifications